

Hack for Humanity Miami: Challenge Guide

Oct 6, 2026 · @Aman

Build what Miami needs next. On Saturday, Oct 10, 2026 , 10 AM to 5 PM at The LAB Miami, you'll pick one of four challenges, build for nearly five hours, and demo a working answer to a real Miami problem.

Hack for Humanity is [The AI Collective's global civic hackathon](#), running in 120 chapters across 50 countries this fall. Miami's edition is also an [MLH Hacktoberfest](#) Hack Day, so open-source AI gets its own prize.

The four challenges

Pick one track as your primary when you submit. Every project can also compete for the MLH Best Open-Source AI Project prize.

Track	The question	You might build
01 Four Forces, One City	How does Miami stay safe and livable as water, air, land and fire push back?	Street-level flood alerts, heat-safety copilots for job sites, storm-ready offline info hubs
02 Wild Miami	How do we protect the animals we share Miami with, and keep neighbors safe too?	Injured-wildlife reporting agents, roadkill hotspot maps, shelter adoption matchers
03 Brain Gain	How does Miami keep the talent it attracts?	Career pathway navigators, AI reskilling coaches, newcomer onboarding guides
04 Data to Action	What does the data say, and what should Miami do about it?	Dashboards, maps, alerts or agents built on open city data

City partner challenge: a public-sector partner may add a challenge and datasets. Details drop before Oct 10, and using them is always optional.

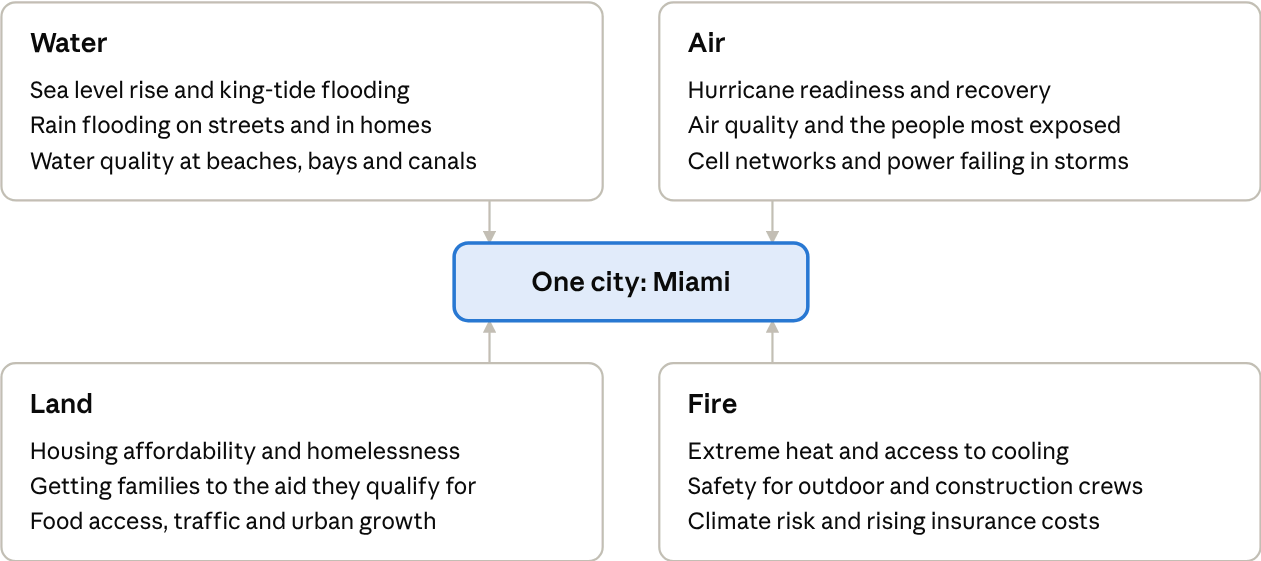
Track 01 - Four Forces, One City

Miami lives with four forces - water, air, land and fire - and each one lands hardest on the neighbors with the fewest options. Pick one force, or combine them, and build for the people living with it.

Start from a person, not a technology: "construction crews in Brickell on a heat-advisory day" beats "a heat app." Climate disaster readiness can cut across all four forces.

Four forces, one city: pick the one your build answers

Each force lands hardest on the neighbors with the fewest options



Track 01 · four forces, one city

Each force lists problems worth solving; the build ideas below show where to start.

Build ideas by force

Force	Build ideas
Water	Street-level flood alerts from tide and rain data; flood-safe routing for commuters and ambulances; a water-quality checker for beaches and canals
Air	A hurricane-prep copilot in English, Spanish and Haitian Creole; an offline-first info hub for when cell networks go down; air-quality alerts for people with asthma

Force	Build ideas
Land	A benefits navigator that matches families to the programs they qualify for; a live shelter-bed and food-pantry finder; a traffic-and-flood commute planner
Fire	Heat-risk alerts for outdoor and construction crews; a cooling-center finder; a climate-risk and home-insurance explainer for homeowners

Guidelines

- Name your force (or forces) and the specific people you serve, in your README and your demo.
- Ground it in Miami: local data, real neighborhoods, and the languages people here speak.
- Show the moment of use: who opens it, when, and what they do next.

Track 02 - Wild Miami

Miami shares its streets with iguanas, ducks, chickens and Everglades wildlife, and too many of them end up hurt, lost or hit by cars. Build tools that protect animals and the people around them.

Challenge	Build ideas
Urban wildlife and invasive species (iguanas, pythons)	A photo-to-species agent that tells residents what the animal is, whether it's dangerous, and who to call
Roadkill	A hotspot map from crash and sighting reports that recommends crossing signs or speed changes
Everglades and endangered species (Florida panthers, manatees, sea turtles)	A sighting and injury reporting tool that routes each case to the right rescue group
Rescue and shelters	A lost-and-found pet matcher, a foster and adoption matchmaker, or a hurricane pet-evacuation planner

Challenge	Build ideas
Cruelty and neglect	A fast, anonymous reporting flow that captures location, photos and urgency

Guidelines

- Animal welfare comes first: never encourage people to chase, handle or harm wild animals themselves - route them to professionals.
- Know who acts on your output (a rescue group, animal services or a resident) and design that handoff.
- Public health counts: projects that reduce human-wildlife conflict are in scope.

Track 03 - Brain Gain

Miami attracts talent; keeping it is the hard part. Build tools that help people grow a career here and stay. This track ties into Hack for Humanity's [global themes](#): job displacement, technical literacy and accessibility.

Challenge	Build ideas
Graduates leaving after school	A career pathway navigator that maps a student's skills to Miami employers, internships and mentors
AI reshaping jobs	A reskilling coach that shows a worker which skills transfer and the shortest path to a growing role
Cost of living	An affordability planner that weighs salary, rent, commute and flood risk by neighborhood
Newcomers without a network	An onboarding guide to Miami's communities, meetups and mentors, matched to someone's goals
Access for everyone	Career tools that work for non-English speakers, people with disabilities, and people without a degree

Guidelines

- Design for one specific person: a graduating senior, a hospitality worker, a relocated engineer.

- Show what would make them stay, and how your tool moves that needle.
- Skip generic job boards: the value is in Miami-specific matching, guidance or community.

Track 04 - Data to Action

Turn data into a decision: find one insight Miami can act on, make it visible, and show the action it unlocks. You don't need a full app - a sharp analysis, a clear visual and a recommended action is a complete submission.

What a complete entry shows

1. **The question** - one Miami problem, stated plainly.
2. **The insight** - one sentence with the number that matters.
3. **The visual** - a chart, map or dashboard anyone can read in 10 seconds.
4. **The action** - who should do what, based on your finding.
5. **The pipeline** - data sources and code in your repo, so judges can reproduce it.

Recommended tools (optional - use what you know)

- **Snowflake** - the [CoCo AI coding agent](#) for exploring data and writing queries, plus free [Marketplace](#) and [sample](#) datasets ([free trial](#)).
- **Gemini API** - including [Gemma 4](#), Google's open-weight model, which also counts toward the open-source AI prize.
- **MCP servers and agents** that let people question data in plain language.

Any topic works: flooding, wildlife, housing, talent. Cite every dataset in your README. City partner datasets, if they land, are optional.

Build requirements

Build it on the day, ship it in public, and demo it live.

Teams

- Solo to 4 people. Solo builders are welcome.
- Want teammates? Say so in your registration, or find the team-matching whiteboard in the lounge at check-in.
- Teams can change until build time starts at 10:45 AM. After that, teams are locked.

What you can bring

- Ideas, research and sketches from before the event: yes.
- Libraries, frameworks, APIs and open-source code: yes.
- Any AI tools, including coding agents: yes. List the ones you used.
- Code or project materials written before the event: no. All project work happens during build time, per the [MLH Standard Hackathon Rules](#).

Open source

- Your code lives in a public GitHub repo with an open-source license (MIT or Apache-2.0 work well) and stays public after the event to remain prize-eligible.
- For the [MLH Best Open-Source AI Project](#) prize, open-source or open-weight AI must be a core part of the project: a Gemma, Llama, Mistral or Qwen model, an open-source model harness, or an agent skill built to the [Agent Skills standard](#). Name the model and link its license in your README.

Submission checklist (in OrganizerHQ, by 3:45 PM)

- ☐ Checked in through OrganizerHQ at the door - required to submit and to win
- ☐ Project name and a short description: the problem, who it helps, how it works
- ☐ Public GitHub repo link with an open-source license
- ☐ Technologies used, including AI models
- ☐ Challenge track selected
- ☐ Demo URL or video (optional)
- ☐ README with your track, data sources, models used, how to run it, and team members

Demo

- 2 minutes per team plus 1 minute of questions. Show the working product, not slides.
- Every team that demos earns an "I Demoed" sticker.

Code of Conduct

- The [MLH Code of Conduct](#) applies throughout the venue and every event activity. Harassment is not tolerated; find any organizer if you have a concern.

Judging rubric

Judges score every demo from 1 to 5 on five criteria, and the weighted total decides the winners. Community impact carries the most weight, because that's the point of the day.

Criterion	Weight	What judges look for
Community impact	30%	A real Miami problem, a specific group of people, and a believable path to helping them
Working build	25%	It runs live and the core flow works end to end. For Data to Action: a reproducible analysis and a visual that answers the question
Use of AI and data	20%	AI and data do real work, not decoration, and judges can verify how
Originality and craft	15%	A fresh angle or thoughtful technical work, built today
Demo clarity	10%	Problem, solution and proof in 2 minutes

How winners are picked

- **Grand Champion:** the highest weighted score overall.
- **Track Champions:** the highest score in each track, excluding the Grand Champion, so five different teams win.
- **MLH Best Open-Source AI Project:** judged separately on MLH's criteria (open-source or open-weight AI at the core, public repo and license, a working demo) and stackable with any other prize.
- **Ties** go to the higher Community impact score. Organizers verify submissions before demos, and rule or Code of Conduct breaks can mean disqualification.

Prize categories

Five teams take home Hack for Humanity titles, one team wins MLH's open-source AI prize, and every builder leaves with swag.

Prize	Who wins	What you get
Grand Champion	Highest overall score across all tracks	Arduino boards for the team
Track Champion (x4)	Top team in each track	MLH Hacktoberfest swag and a spotlight on our community channels

Prize	Who wins	What you get
MLH Best Open-Source AI Project	Best project with open-source or open-weight AI at its core	A DEV Badge for every team member, plus MLH+DEV swag while supplies last
City Partner Challenge	Announcing soon	Details before Oct 10
Every builder	Everyone who checks in	Hacktoberfest swag (T-shirts, stickers, postcards while supplies last), plus an "I Demoed" sticker when your team presents

All winning teams may be invited to join a grant-funded project to commercialize what they built, and may be showcased in The AI Collective's global newsletter to 300k+ subscribers ([event page](#)).

Schedule

Build time runs 10:45 AM to 3:30 PM, and submissions close at 3:45 PM sharp, so submit early. All times Eastern, per the [event page](#).

Time	What happens
10:00-10:20 AM	Check-in through OrganizerHQ, swag, team matching in the lounge
10:20-10:45 AM	Welcome, challenge briefing and MLH Hacktoberfest intro
10:45 AM	Build time starts and teams lock
12:30-2:00 PM	Food available - eat when it suits your team
3:30 PM	Build time ends
3:30-3:45 PM	Submit in OrganizerHQ (deadline 3:45 PM)
3:45-4:30 PM	Demos
4:30-4:45 PM	Judging

Time	What happens
4:45-5:00 PM	Winners and closing

Resources

Optional starting points, gathered so you spend build time building, not searching. Bring your own data too.

Resource	Use it for	Tracks
Miami-Dade County Open Data Hub	County-wide datasets and maps	All
City of Miami Data Explorer	City-level datasets and maps	All
NOAA Tides & Currents	Tide levels and sea level trends	01, 04
National Hurricane Center	Storm tracks, advisories and past storms	01, 04
AirNow API	Current and forecast air quality	01, 04
OpenFEMA	Disaster declarations, flood insurance claims, assistance data	01, 04
U.S. Census data	Income, housing, language and demographics by area	01, 03, 04
BLS data	Jobs, wages and employment trends by metro area	03, 04
O*NET OnLine	Skills, tasks and career paths by occupation	03
iNaturalist	Community wildlife sightings with photos and locations	02, 04
GBIF	Species occurrence data	02, 04
Gemma 4 on the Gemini API	Open-weight multimodal model (quickstart)	All
Snowflake CoCo	AI coding agent for data work (free trial)	04
Hugging Face and Ollama	Open-weight models in the cloud or on your laptop	All

Resource	Use it for	Tracks
Agent Skills standard	The spec for packaging open agent skills	All

Official links: [Event page](#) · [MLH Code of Conduct](#) · [MLH Standard Hackathon Rules](#) · [MLH Best Open-Source AI Project rules](#) · [Hack for Humanity announcement](#)